Toledo - Driver Pay Optimization

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1. Context & Key Question

Route: Toledo Airport ? Downtown

Current Fare: $25 (Cannot exceed this rate)

Driver Pay: Historically $19 ? 60% match rate; at $22 ? 93% match rate

Key Question: How much more or less should we pay drivers per trip (and changing Lyft's take from the $25 fare) to maximize Lyft net revenue for the next 12 months?

Recommendation: Pay drivers $20.87 per trip to maximize net revenue in the steady-state model. Future adjustments may be necessary if real-world conditions (e.g., seasonality, changing churn) deviate from our baseline assumptions. See model here.

2. Overview of the Analysis

We built a steady-state financial model capturing:

Parameter

Value/Range

Notes

Rider Fare

$25

Cannot exceed prevailing market rate.

Driver Pay ($x)

$19 - $22 (tested)

Decision variable in model: test any $19 ? x ? $25).

Match Rate - M(x)

60% @ $19 ? 93% @ $22

Assumed linear: M(x)=0.11?x?1.49 (see appendix for info)

Monthly Rider Base

10,000

Maintained by replacing lost riders

Rider Churn

10% (if matched)

Overall = 0.10×M(x) + 0.33×(1?M(x))

33% (if no match)

Rider CAC

$15 per new rider

Avg cost to acquire each replaced/lost rider.

Driver Churn

5% per month

Baseline assumption.

Driver Output

100 rides/driver/month

Used to calculate # of drivers needed.

Driver CAC

$500 per new driver

Avg cost to recruit each replaced driver.

Why these assumptions?

We lack the ability to raise fares above $25.

Driver pay directly affects driver supply ? match rate.

Rider churn heavily influences CAC if they fail to find a driver.

Steady-state ensures each month's rider/driver count is re-topped to the same base.

Unit Economics repeated monthly for 12 months, with a "top up" to keep 10,000 active riders and enough drivers to fulfill rides.

Net Revenue each month: Net Revenue (Month) = (Fare - x) * (Completed Rides) - (Rider CAC) - (Driver CAC)

Summed over 12 months = 12 * Net Revenue (Month)

3. Key Model Mechanics: Steady State

Element

Formula / Calculation

Completed Rides

10,000 * M(x)

Ride Margin (pre-CAC)

(25 - x) * [10,000 * M(x)]

Rider CAC (monthly)

Riders Lost = 10,000 * [0.33 - 0.23 * M(x)]

CAC = (Riders Lost) × $15.

Driver CAC (monthly)

Drivers Needed = (10,000 * M(x)) / 100

5% churn ? new drivers each month.

CAC = (New Drivers) × $500.

Monthly Net

Ride Margin (pre-CAC) - Rider CAC - Driver CAC

Annual Net

12 * (Monthly Net)

Note: We iterated driver pay from $19 to $22 in $0.01 steps using a solver.

4. Findings & Core Recommendation

4.1 Model output

Driver Pay

Match Rate

Completed rides (mo)

Driver Churn %

Annual Net

$19

~60%

~6,000

19.2%

~$68k

$20.87

~79%

~7,900

14.8%

~$115k

$22

~93%

~9,300

11.6%

~$96k

Lower Pay ($19): Higher per-ride margin but limited coverage (60%), leading to heavy rider churn and thus high rider CAC. Annual net ~$68k.

Higher Pay ($22): Great coverage (93%) but margin drops to $3/ride, capping net at ~$96k/year.

Sweet Spot ($20.87): Coverage + margin both reasonably strong, maximizing 12-month net at ~$115k.

At $19

Coverage: Only ~60% of requests get matched, yielding ~6,000 completed rides per month.

Effect: Although Lyft's margin per ride is high ($6 per trip), the large proportion of failed requests increases rider churn (33% among those who fail).

Result: Heavy rider churn leads to higher monthly marketing costs (CAC) to re-acquire lost users, capping annual net at about $68k.

At $22

Coverage: Improves significantly to ~93%, or ~9,300 rides per month.

Effect: Rider churn plummets, but Lyft's margin shrinks to only $3 per ride.

Result: Despite high coverage, the low per-ride margin limits annual net to roughly $96k.

At $20.87 (the "Sweet Spot")

Coverage: A balanced ~79%, translating to ~7,900 completed rides per month.

Margin: $25 - $20.87 = $4.13, still meaningfully higher than $3 at the $22 level.

Effect: Keeps rider churn at a moderate level while sustaining a decent margin per ride.

Result: Maximizes the overall 12-month net at ~$115k, representing a significant jump from $19 and $22 scenarios (up to $47k more than the $19 option, and $19k more than $22).

4.2 Conclusion & Action

Implement $20.87 Driver Pay

Given the steady-state math, this rate optimizes coverage vs. churn trade-offs and drives the highest annual net (about $115k).

Monthly net under this rate is roughly $9.6k, a clear improvement over the $19 or $22 alternatives.

Why This Matters

Too Low ($19): Lyft's margin is higher, but coverage collapse triggers outsized rider churn and high CAC.

Too High ($22): Coverage soars, yet thin margins reduce total net.

Balanced ($20.87): Enough driver incentive to keep coverage high, but not so high that margins erode.

Next Steps / Actions

Adopt a $20.87 pay structure for the next 12 months.

Monitor real-world churn rates, driver supply, and marketing spend; adjust if data diverges from steady-state assumptions.

Consider future refinements if seasonal demand or marketing constraints become relevant.

Dynamic Approach if we suspect monthly fluctuations (seasonality, marketing constraints).

Driver Churn Sensitivity if paying more reduces churn below 5%.

Appendix: Steady State vs. Dynamic

Aspect

Steady State

Dynamic Month-by-Month

Approach

Repeats the same monthly patterns; "top up"

Each month evolves with real churn, changing coverage

Simplicity

Quick to compare pay levels in equilibrium

More inputs; deeper realism (seasonality, budgets)

When to Use

High-level pricing decisions

Operational forecasting, multi-phase pay changes

Appendix: Getting Match rate formula

? M(x) = 0.11 * x - 1.49

When driver pay x = 19, match rate M(x) = 0.60

When driver pay x = 22, match rate M(x) = 0.93

Let's calculate the slope:

Slope = (change in match rate) / (change in x)

= (0.93 - 0.60) / (22 - 19)

= 0.33 / 3

= 0.11

M(x) = slope * x + intercept ? using x = 19, M = 0.60

0.60 = (0.11 * 19) + intercept

0.60 = 2.09 + intercept

intercept = 0.60 - 2.09

intercept = -1.49